

Sai Harshit Balaji

Aspiring Robotics Engineer — Robotics, Autonomous Systems, Embedded Software & IoT
Cham, Bavaria, Germany | (+49)15510532825 | sai.harshitbalaji@gmail.com | DOB: 24.02.2002, India
[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Technische Hochschule Deggendorf

Master of Engineering in Intelligent Robotics

Cham, Germany

Mar 2026 – Present

- Focus Areas: Robot Dynamics, Advanced Control Engineering, Computer Vision, Embedded Systems, ROS

BMS Institute of Technology and Management

Bachelor of Engineering in Computer Science and Engineering | CGPA: 7.89/10

Bengaluru, India

Aug 2020 – Jul 2024

- Strong foundations in: Programming, Algorithms, AI/Machine Learning, Software Design and Development

WORK EXPERIENCE

Academic Research Projects

Jul 2025 – Mar 2026

BMS Institute of Technology & Management

Bengaluru, India

- Conducted research in **federated learning**, secure IoT systems, and intelligent anomaly detection under the guidance of Dr. Arun Kumar B R. Authored and presented 3 research papers focused on privacy-preserving IoT security, cyber-physical anomaly intelligence, and intelligent monitoring systems.
- Designed a proof-of-concept **IoT-enabled blood bank logistics and monitoring platform** integrating anomaly awareness, auditability, and intelligent tracking concepts.
- Designed and developed a custom 5-DOF robotic arm platform with stereo vision capability and dedicated embedded processors for robotic control and perception workloads, targeting ROS2 Humble integration. Done under the guidance and support of Dr. Ambika G N

Graduate Engineer Trainee

Jul 2024 – Jul 2025

Carl Zeiss AG

Bengaluru, India

- Designed and developed a full-stack lab infrastructure management platform using **FastAPI, React, SQL Server, and Azure** for real-time monitoring and management of connected devices across multiple R&D labs.
- Prototyped a **Windows-based diagnostics tool** in C# for ZEISS medical and industrial devices, automating structured health checks and reducing device service time by an estimated 20–25%.
- Contributed to architecture and feasibility discussions for an AI-assisted predictive diagnostics framework for hardware systems.
- Collaborated across software, QA, and engineering teams in Agile workflows while developing internal automation and infrastructure tooling.

Software Development & QA Intern

Jan 2024 – Jun 2024

Carl Zeiss AG

Bengaluru, India

- Contributed to the development and Azure deployment of an internal feature-store platform supporting machine learning workflows.
- Built a **React-based CI monitoring dashboard** for visualization of build and deployment activity, reducing manual monitoring overhead.
- Maintained and optimized automation pipelines and backend tooling using **C#**, Selenium, and Azure DevOps.

Robotics R&D Intern

Aug 2023 – Nov 2023

*Defence Research and Development Organisation (DRDO)
Centre for Artificial Intelligence and Robotics (CAIR)*

Bengaluru, India

- Integrated **MoveIt2** with **PlanSys2** in **ROS2 Humble** using Python and C++ to enable autonomous task execution on a 6-DOF robotic arm platform.
- Benchmarked and optimized **OMPL motion planners**; tuned RRTConnect to achieve RRT*-like path consistency with significantly reduced runtime.
- Performed simulation-based validation and optimization of robotic planning and execution workflows in restricted research environments.

PROFESSIONAL SKILLS

Robotics: ROS2, MoveIt2, PlanSys2, RViz, Gazebo, OMPL, Motion Planning

Programming Languages: Python, C++, C#, JavaScript, SQL, Java

Embedded & IoT: Raspberry Pi, ESP32, IoT Systems, Sensor Integration, Edge Systems

Backend & Software Systems: FastAPI, ReactJS, REST APIs, SQLAlchemy, RBAC, Distributed Systems Concepts

Tools & Platforms: Linux, Git, Azure, Docker (basic), Postman, Selenium, Azure DevOps

Engineering Practices: Object-Oriented Programming, Agile Development, System Design, Automation

Soft Skills: Systems Thinking, Analytical Problem Solving, Technical Communication, Cross-Functional Collaboration, Independent Learning

PROJECTS

HB-A5-01: 5-DOF Robotic Arm | *Embedded Systems, Raspberry Pi, Robotics*

Aug 2025 – Present

- Designed and assembled a custom 5-DOF robotic manipulator with a reach of approximately 70 cm for robotics experimentation and autonomous manipulation research.
- Integrated dual processing units comprising a Raspberry Pi 4 for robot control and a StereoPi v2 system with Compute Module 4 for stereo vision workloads.
- Implemented microcontroller-based joint control and integrated an IMX219-83 stereo camera; ROS2-based perception and motion planning capabilities are currently under development.

Autonomous Voice Controlled Arm Robot | *ROS2, MoveIt2, PlanSys2, YOLOv8*

Sep 2023 – May 2025

- Developed a robotic arm system capable of autonomous task execution through voice commands using ROS2-based planning and control pipelines.
- Integrated **PlanSys2** for symbolic task planning and **MoveIt2** for motion planning and trajectory execution in RViz and Gazebo simulation environments.
- Implemented object detection using **YOLOv8** for real-time object identification and interaction workflows.
- Designed as an assistive robotics system for elderly and bedridden patient support scenarios.

PUBLICATIONS

Fed-AE: Privacy-Preserving and Zero-Day Intrusion Detection for IoT Edge Networks

- Presented at the **IEEE Madhya Pradesh Section Conference (MPCON 2026)**; published in IEEE Xplore.

Unified Operational-Cyber Anomaly Intelligence for Risk-Aware IoT Blood Bank Cold-Chain Monitoring

- Presented at the **International Conference on Computing Systems and Intelligent Applications (ComSIA 2026)**; currently under process for publication in the **Springer LNNS** series.

IoT-Enabled Geofence and RFID Passkey System for Secure Blood Bag Handover in Cold Chain Logistics

- Presented at **IMPACT 2025 – Intelligent Machine Processes and AI-Centric Technologies**; currently under process for publication in the **Springer LNNS** series.

LANGUAGES

English: Full Professional Proficiency | **German:** A1 (currently pursuing A2/B1) | **Telugu:** Native Language

ACHIEVEMENTS

- Awarded **Best Paper Award** at the International Conference on Computing Systems and Intelligent Applications (ComSIA 2026) for research on cyber-physical anomaly intelligence in IoT blood bank cold-chain systems.
- Coordinated and led a **5-day hands-on workshop on ROS and Robotics** for undergraduate students, introducing ROS fundamentals, robot simulation, and autonomous systems concepts.
- Served as **Student Placement Coordinator**, facilitating career events, supporting student-industry engagement, and coordinating activities between students and the Training & Placement Cell.

INTERESTS

Technical: Robotics, Autonomous Systems, Embedded Electronics, Open-Source Software, Arduino & ESP32 Projects

Personal: Hiking, Gardening, Chess, Football, Badminton, Space, Geography, Indian History & Philosophy